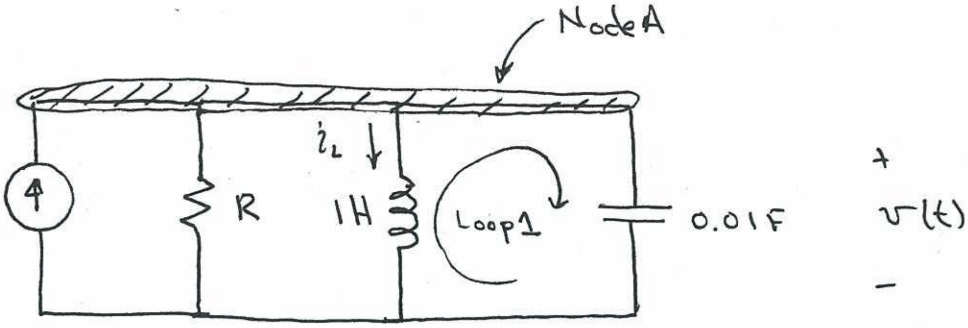
8.1



(a) Define the current through the inductor $= i_L(t)$

KCL at node A: $i(t) = \dfrac{v(t)}{R} + i_L(t) + (0.01F)\dfrac{dv(t)}{dt}$ (*)

KVL, loop 1: $v(t) = (1H)\dfrac{di_L(t)}{dt}$ (**)

Differentiate equation (*) with respect to time:

$$\frac{di(t)}{dt} = \frac{1}{R}\frac{dv(t)}{dt} + \frac{di_L(t)}{dt} + 0.01\frac{d^2v(t)}{dt}$$

substitute equation (**):

$$\boxed{100\frac{di(t)}{dt} = \frac{100}{R}\frac{dv(t)}{dt} + 100\,v(t) + \frac{d^2v(t)}{dt}}$$

(b) From above, $\omega_n^2 = 100 \Rightarrow \underline{\omega_n = 10\ \text{rad/sec}}$

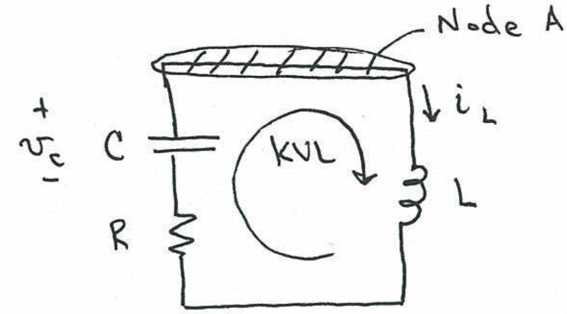
$2\xi\omega_n = \dfrac{100}{R} \Rightarrow 2(1)(10\ \text{rad/s}) = \dfrac{100}{R} \Rightarrow \underline{R = 5}$

$\xi = 1$, for critical damping

$$\boxed{R = 5\Omega}$$

8.2

After switch opens, circuit becomes:



Define two unknowns, v_c & i_L

(a) KCL, Node A: $i_L = -C\dfrac{dv_c}{dt}$

KVL: $v_c + RC\dfrac{dv_c}{dt} = L\dfrac{di_L}{dt}$

$$\frac{d^2v_c}{dt^2} + \frac{R}{L}\frac{dv_c}{dt} + \frac{1}{LC}v_c = 0$$

With $R = 2\Omega$, $C = 1\mu F$, $L = 1mH$:

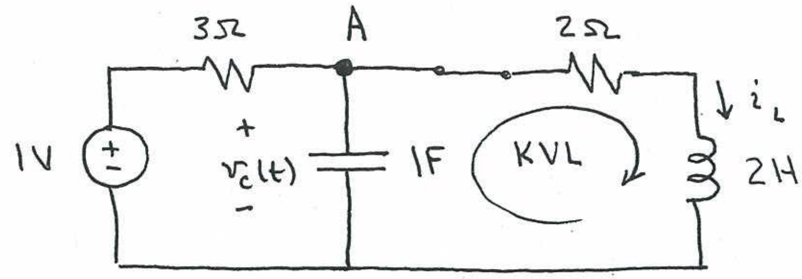
$$\boxed{\frac{d^2v_c}{dt^2} + 2000\frac{dv_c}{dt} + 1\times10^9 v_c = 0}$$

(b) $\omega_n = \sqrt{1\times10^9} = 3.16\times10^4$ rad/sec

$\xi = \dfrac{2000}{2\omega_n} = 0.0316$

$\xi < 1 \Rightarrow$ $\boxed{\text{system is underdamped}}$

8.3



(a) KCL at A: $\dfrac{1-v_c}{3\Omega} = (1F)\dfrac{dv_c}{dt} + i_L \Rightarrow 1 - v_c = 3\dfrac{dv_c}{dt} + 3i_L$ (*)

KVL: $v_c = (2\Omega)\, i_L + (2H)\dfrac{di_L}{dt}$ (**)

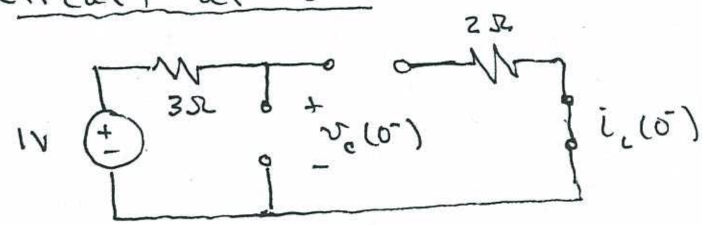
Substitute (**) into (*):

$$1 = \left[2i_L + 2\frac{di_L}{dt}\right] + 3\left[2\frac{di_L}{dt} + 2\frac{d^2 i_L}{dt^2}\right] + 3i_L$$

$$\boxed{1 = 5i_L + 8\frac{di_L}{dt} + 6\frac{d^2 i_L}{dt^2}}$$

(b) Initial condition

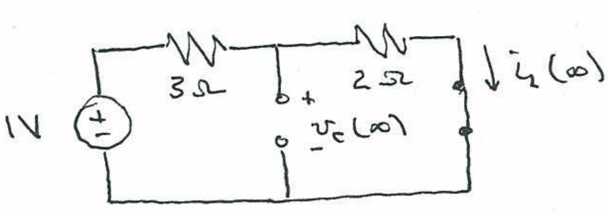
Circuit at $t=0^-$:



$$\boxed{\begin{gathered} v_c(0) = 1V \\ i_L(0) = 0A \end{gathered}}$$

Final Condition:

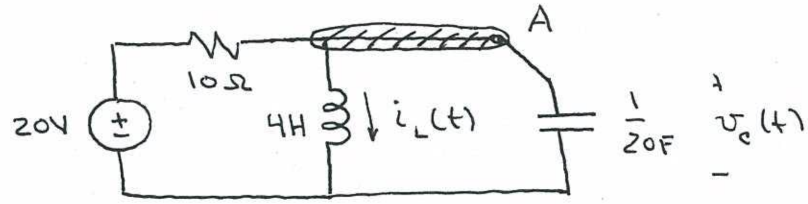
Circuit at $t \Rightarrow \infty$:



$$\boxed{\begin{gathered} v_c(\infty) = \frac{2\Omega}{2\Omega + 3\Omega} = \frac{2}{5}\,V \\ i_L(\infty) = \frac{1V}{5\Omega} = \frac{1}{5}\,A \end{gathered}}$$

8.4

(a) For $t > 0$, circuit becomes:



KVL, right-most loop: $v_c = 4 \dfrac{di_L}{dt}$

KCL, node A: $\dfrac{20 - v_c}{10\Omega} = i_L + \dfrac{1}{20}\dfrac{dv_c}{dt}$

Combining:

$$2 - \frac{4 di_L}{dt} = i_L + \frac{1}{20}\left(4\frac{d^2 i_L}{dt^2}\right)$$

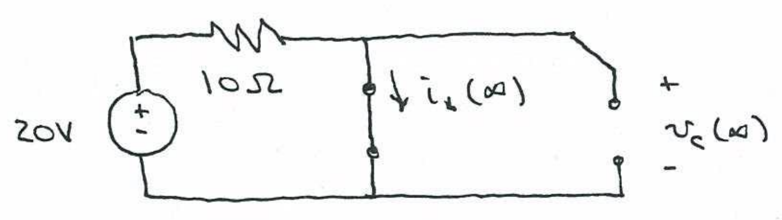
$$2 = \frac{1}{5}\frac{d^2 i_L}{dt^2} + 4\frac{di_L}{dt} + i_L \Rightarrow \boxed{\frac{d^2 i_L}{dt^2} + 20\frac{di_L}{dt} + 5 i_L = 10}$$

(b)

Initial condition:

$$\boxed{\begin{aligned} i_L(0) &= \frac{20V}{10\Omega} = 2A \\ v_c(0) &= 4V \end{aligned}}$$

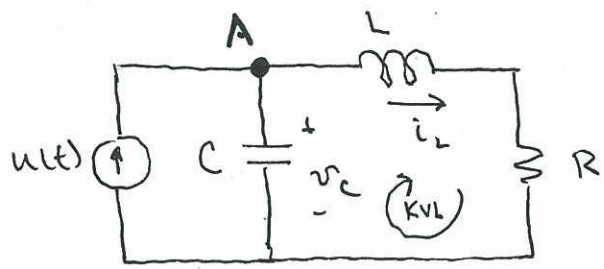
Final condition:



$$\boxed{\begin{aligned} v_c(t \to \infty) &= 0V \\ i_L(t \to \infty) &= \frac{20V}{10\Omega} = 2A \end{aligned}}$$

8.5

For generic R, L, C: (we'll plug in specific values later)



(a) Defining unknowns v_c, i_L as shown & applying KVL & KCL:

KCL at A: $u = C\dfrac{dv_c}{dt} + i_L$

KVL, right loop: $v_c = L\dfrac{di_L}{dt} + Ri_L$

$$u = \frac{d^2 i_L}{dt^2} + \frac{R}{L}\frac{di_L}{dt} + \frac{1}{LC}i_L$$

with $R = 4\Omega$, $L = 1H$, $C = 0.01F$ & noting that $i = i_L$:

$$\boxed{100u = \frac{d^2 i_L}{dt^2} + 4\frac{di_L}{dt} + 100 i_L}$$

(b) From coefficients of differential equation of part (a):

$$\left.\begin{aligned} \omega_n &= \sqrt{100} = 10 \ \text{rad/sec} \\ 2\xi\omega_n &= 4 \Rightarrow 2\xi(10) = 4 \Rightarrow \xi = \frac{4}{20} \end{aligned}\right\} \Rightarrow \boxed{\begin{aligned} \omega_n &= 10 \ \text{rad/s} \\ \xi &= 0.2 \end{aligned}}$$

continued

8.5 (cont'd)

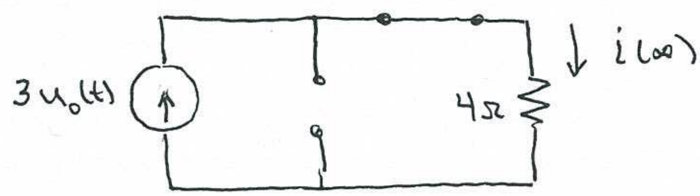
(c)

$$M_p \approx 1 - \frac{\zeta}{0.6} = 1 - \frac{0.2}{0.6}$$ $\zeta = 0.2$ from part (a)

$\underline{M_p = 0.67}$

steady-state response:

Circuit as $t \to \infty$, with $u(t) = 3u_o(t)$ A:



$\Rightarrow \underline{i(\infty) = 3\text{ A}}$

Maximum value:

$$i_{max} = 3\text{A} + 0.67(3\text{A})$$

$$\boxed{i_{max} = 5\text{ A}}$$

8.6

$$\frac{d^2 i(t)}{dt^2} + 6\frac{di(t)}{dt} + 100\, i(t) = 50\, u_0(t)$$

(a) $\omega_n = \sqrt{100} \Rightarrow \boxed{\omega_n = 10 \text{ rad/sec}}$

$2\zeta\omega_n = 6 \Rightarrow \zeta = \frac{6}{2(10)} \Rightarrow \boxed{\zeta = 0.3}$

$\omega_d = \omega_n\sqrt{1-\zeta^2} = 10\sqrt{1-0.3^2} \Rightarrow \boxed{\omega_d = 9.54 \text{ rad/s}}$

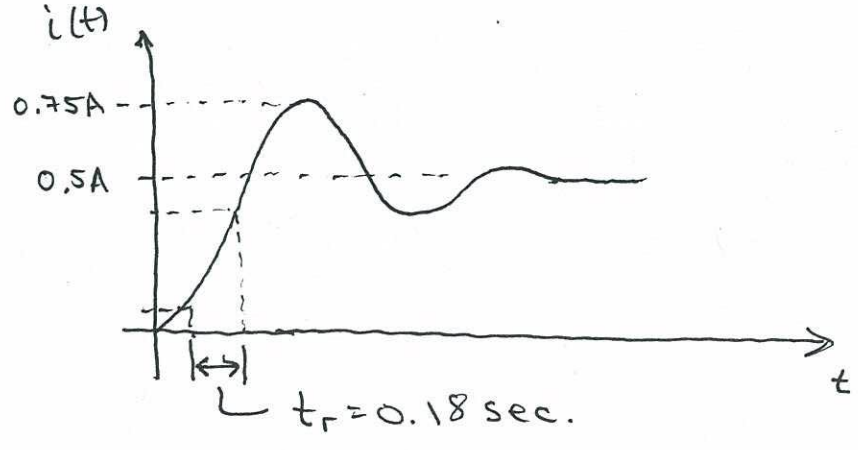
(b) Determine steady-state response, t_r, maximum value:

steady-state response (from diff. eqn):

$$\underbrace{\cancel{\frac{d^2 i_{ss}}{dt^2}}}_{0} + 6\underbrace{\cancel{\frac{di_{ss}}{dt}}}_{0} + 100\, i_{ss} = 50 \Rightarrow i_{ss} = \frac{50}{100} = 0.5A$$

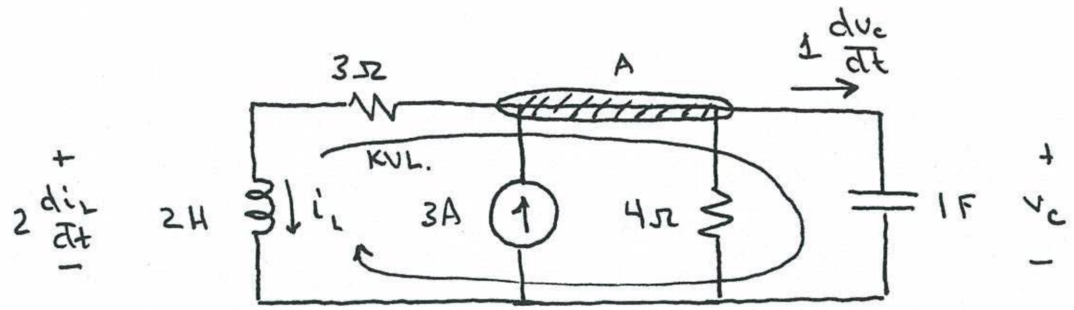
$$M_p \cong 1 - \frac{\zeta}{0.6} = 1 - \frac{0.3}{0.6} = 0.5 \Rightarrow i_{peak} = 0.75\,A$$

$$t_r \cong \frac{1.8}{\omega_n} = 0.18 \text{ sec}$$



8.7

(a) Circuit for $t > 0$:



KCL, node A: $3A = i_L + \frac{v_c}{4\Omega} + \frac{dv_c}{dt}$

KVL, loop shown: $2\frac{di_L}{dt} + 3i_L = v_c$

$$\frac{dv_c}{dt} = 2\frac{d^2 i_L}{dt^2} + 3\frac{di_L}{dt}$$

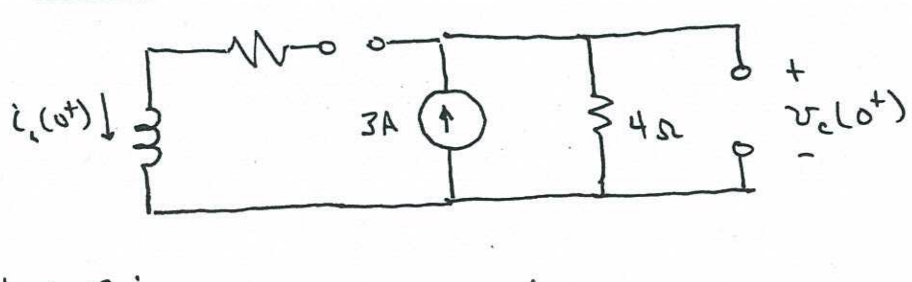
$\frac{d}{dt}$

Substituting:

$$3A = i_L + \frac{1}{4}\left[2\frac{di_L}{dt} + 3i_L\right] + \left[2\frac{d^2 i_L}{dt^2} + 3\frac{di_L}{dt}\right]$$

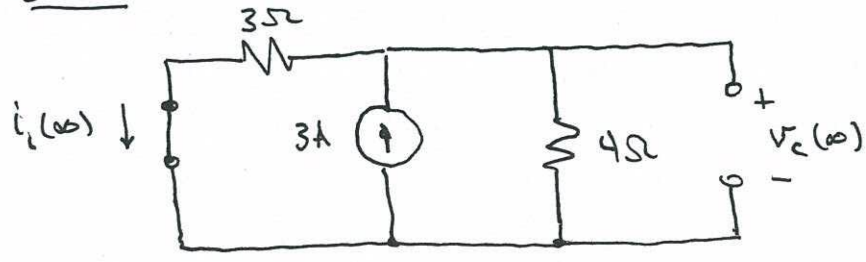
$$\boxed{2\frac{d^2 i}{dt^2} + \frac{7}{2}\frac{di_L}{dt} + \frac{7}{4}i_L = 3}$$

(b) $t = 0^-$:



$$\boxed{\begin{array}{l} i_L(0) = 0\text{ A} \\ v_c(0) = 12\text{ V} \end{array}}$$

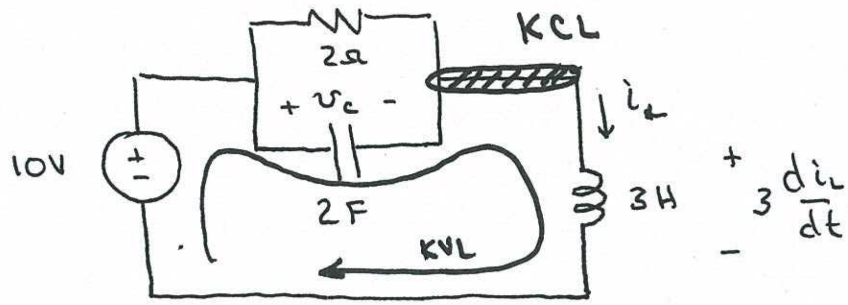
$t \to \infty$:



$$\boxed{\begin{array}{l} i_L(\infty) = \frac{12}{7}\text{ A} \\ v_c(\infty) = \frac{36}{7}\text{ V} \end{array}}$$

8.8

(a) Circuit for $t>0$



KVL as shown: $10V = v_c + 3\frac{di_L}{dt}$

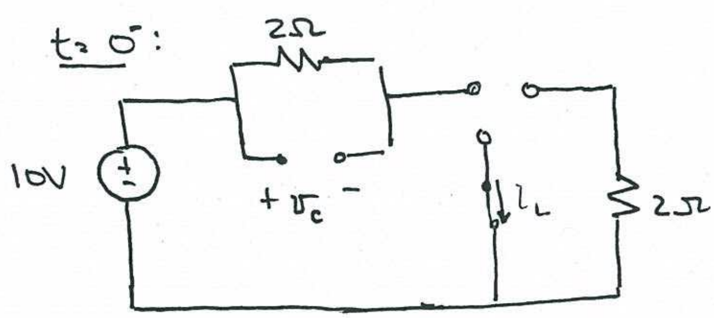
KCL as shown: $\frac{v_c}{2\Omega} + (2F)\frac{dv_c}{dt} = i_L \Rightarrow \frac{di_L}{dt} = 2\frac{d^2v_c}{dt^2} + \frac{1}{2}\frac{dv_c}{dt}$

Substituting:

$$10V = v_c + 3\left[2\frac{d^2v_c}{dt^2} + \frac{1}{2}\frac{dv_c}{dt}\right]$$

$$\boxed{10 = v_c + \frac{3}{2}\frac{dv_c}{dt} + 2\frac{d^2v_c}{dt^2}}$$

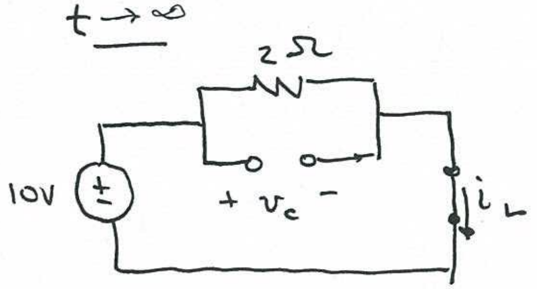
(b) $t=0^-$:



$v_c(0^+) = v_c(0^-) = 10V\left(\frac{2\Omega}{2\Omega+2\Omega}\right)$

$$\boxed{\begin{array}{l} v_c(0^+) = 5V \\ i_L(0^+) = 0A \end{array}}$$

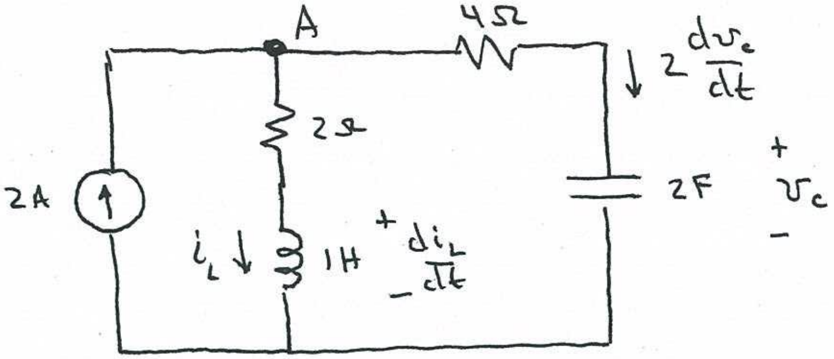
$t\to\infty$



$$\boxed{\begin{array}{l} v_c(t\to\infty) = 10V \\ i_L(t\to\infty) = \frac{10V}{2\Omega} = 5A \end{array}}$$

8.9

(a) Circuit for $t>0$:



KCL at A: $2A = i_L + 2\frac{dv_c}{dt}$ (*)

KVL, right loop: $\frac{di_L}{dt} + 2i_L = 4\Omega\left(2\frac{dv_c}{dt}\right) + v_c$ (**)

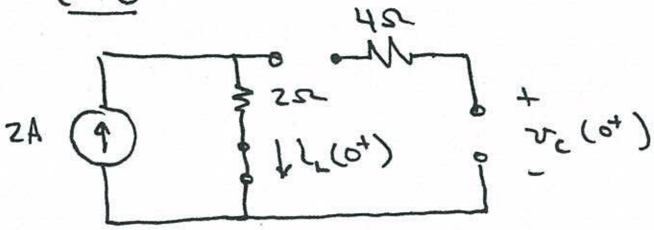
From (*), $i_L = 2 - 2\frac{dv_c}{dt}$ & $\frac{di_L}{dt} = -2\frac{d^2v_c}{dt^2}$

Substituting into (**):

$$-2\frac{d^2v_c}{dt^2} + 2\left[2 - 2\frac{dv_c}{dt}\right] = 8\frac{dv_c}{dt} + v_c$$

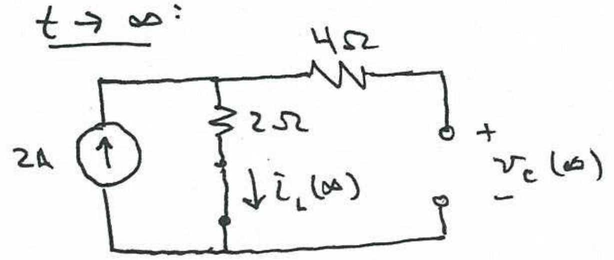
$$\boxed{4 = 2\frac{d^2v_c}{dt^2} + 12\frac{dv_c}{dt} + v_c}$$

(b) $t<0$:



$$\boxed{\begin{aligned} i_L(0) &= 2A \\ v_c(0) &= 0V \end{aligned}}$$

$t\to\infty$:



$$\boxed{\begin{aligned} i_L(\infty) &= 2A \\ v_c(\infty) &= 4V \end{aligned}}$$

8.10

Put differential equation in standard form (divide by 2):

$$\frac{d^2 v_{OUT}}{dt^2} + 2\frac{dv_{OUT}}{dt} + 25\, v_{OUT} = 50\, v_{in}$$

$\omega_n^2 = 25 \Rightarrow \omega_n = 5$ rad/sec

$2\zeta\omega_n = 2 \Rightarrow \zeta = \frac{1}{\omega_n} = 0.2$

$$M_p \approx 1 - \frac{\zeta}{0.6} = 0.67$$

constant input $\Rightarrow \frac{dv_{OUT}}{dt} = 0, \ \frac{d^2 v_{OUT}}{dt^2} = 0$ as $t \to \infty$

$$25\, v_{OUT}(t\to\infty) = 50\, v_{in}(t\to\infty)$$

$$v_{OUT}(t\to\infty) = \frac{50}{25} v_{in}(t\to\infty) = \frac{50}{25}(2V)$$

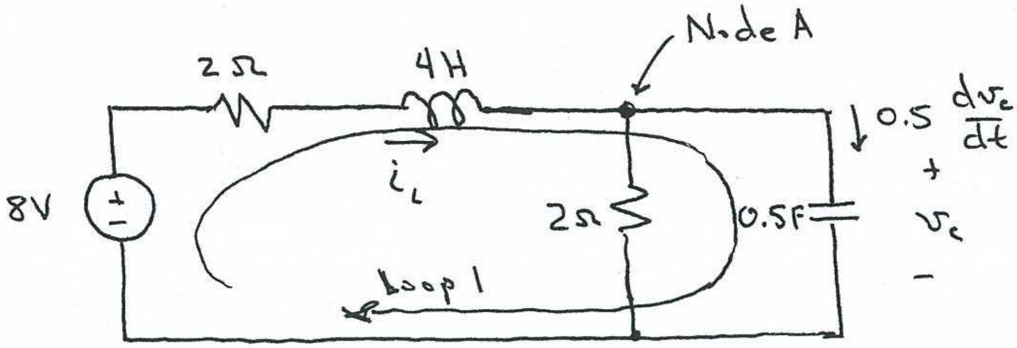
$$v_{OUT}(t\to\infty) = 4V$$

$$v_{OUT}\Big|_{max} = v_{OUT}(t\to\infty)\left[1+M_p\right] = 4V\left[1+0.67\right]$$

$$\boxed{v_{OUT}\Big|_{max} = 6.67\,V}$$

8.11

Circuit for $t > 0$:



KVL, loop 1: $-8V + (2\Omega) i_L + (4H)\frac{di_L}{dt} + v_c = 0$

KCL, Node A: $i_L = \frac{v_c}{2\Omega} + \frac{1}{2}\frac{dv_c}{dt}$

$$\downarrow \frac{d}{dt}$$

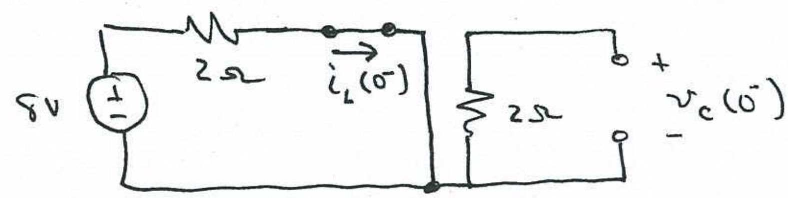
$$\frac{di_L}{dt} = \frac{1}{2}\frac{dv_c}{dt} + \frac{1}{2}\frac{d^2 v_c}{dt^2}$$

Substituting:

$$8 = 2\left[\frac{v_c}{2} + \frac{1}{2}\frac{dv_c}{dt}\right] + 4\left[\frac{1}{2}\frac{dv_c}{dt} + \frac{1}{2}\frac{d^2 v_c}{dt^2}\right]$$

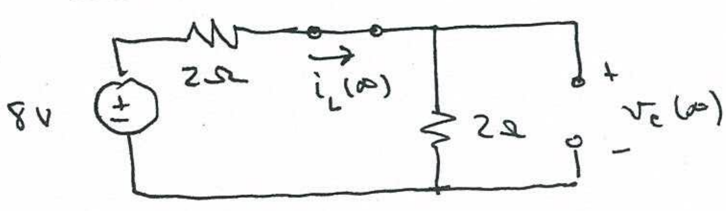
$$\boxed{8 = 2 v_c + 3\frac{dv_c}{dt} + 2\frac{d^2 v_c}{dt^2}}$$

(b) $t = 0^-$:



$$\boxed{\begin{array}{l} v_c(t=0^+) = v_c(0^-) = 0V \\ i_L(t=0^+) = i_L(0^-) = 4A \end{array}}$$

$t \to \infty$:



$$\boxed{\begin{array}{l} v_c(t\to\infty) = 4V \\ i_L(t\to\infty) = 2A \end{array}}$$